

1 Optimization problem

Balancing a disaggregated table, respecting constraints of coherence to original table and including exogenous information.

$$\begin{aligned}
& \min_{Z,Y,V} \sum_{rf,rt} \left(\sum_{sf,st \in S_N} H(Z, Z^0) + \sum_{sf \in S_S, st \in S_P} H(Z, Z^0) \right) + \sum_{f,st,rt} H(V, V^0) \\
& \text{s. t.} \quad \sum_{sf \in S_{PN}} Z\alpha(sf, sf') = Zold \quad \forall sf' \in S'_P, \forall st \in S_s, \forall rf \in R, \forall rt \in R \\
& \quad \sum_{sf \in S_{PN}} \sum_{st \in S_{PN}} Z\alpha(sf, sf')\alpha(st, st') = Zold \quad \forall sf' \in S'_P, \forall st' \in S'_P, \forall rf \in R, \forall rt \in R \\
& \quad \sum_{st \in S_{PN}} Z\alpha(st, st') = Zold \quad \forall sf \in S_S, \forall st' \in S'_P, \forall rf \in R, \forall rt \in R \\
& \quad \sum_{sf \in S_{PN}} \sum_{st \in S_{PN}} Z\alpha(sf, sf')\alpha(st, st') = Zold \quad \forall sf' \in S'_P, \forall st' \in S'_P, \forall rf \in R, \forall rt \in R \\
& \quad \sum_{sf \in S_{PN}} Y\alpha(sf, sf') = Yold \quad \forall sf' \in S'_P, \forall c \in C, \forall rf \in R, \forall rt \in R \\
& \quad \sum_{st \in S_{PN}} V\alpha(st, st') = Vold \quad \forall st' \in S'_P, \forall f \in F, \forall rt \in R \\
& \quad \sum_{rt \in R} \left(\sum_{st \in S} Z + \sum_{c \in C} Y \right) = X(sf, rf) \quad \forall sf \in S_{PN}, \forall rf \in R \\
& \quad \sum_{rf \in R} \left(\sum_{sf \in S} Z + \sum_{f \in F} V \right) = X(st, rt) \quad \forall st \in S_N, \forall rt \in R \\
& \quad \sum_{sf \in S} X\alpha(sf, sf') = Xold(sf', rf) \quad \forall sf' \in S_P, \forall rf \in R \\
& \quad \sum_{rt \in R} \left[\left(\sum_{st \in S} Z + \sum_{c \in C} Y \right) \beta(rf, rt) \right] = T \quad \forall sf \in S_N, \forall rf \in R
\end{aligned} \tag{1}$$

Definition of operators

$$\begin{aligned}
H(x, y) &= (x + \epsilon) \ln[(x + \epsilon)/(y + \epsilon)] & \text{where } \epsilon &= 1e^{-6} \\
\alpha(sf, sf') &= 1 \text{ if } sf' \text{ is parent of } sf, \text{ else } 0 \\
\beta(rf, rt) &= 1 \text{ if } rf \neq rt, \text{ else } 0
\end{aligned} \tag{2}$$